I-MAP: LOGBOOK (AI USAGE)

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LLM request

⚠ Claude was used for this prompt.

Is there a way that I can attach a non-static version of _itWheel? Since I'm using the class, that it is part of, 4 times in my sketch.

void Encoder::_itWheel() { this->_wheel_counter++; }

I currently get this error
lib/encoder/encoder.cpp: In member function 'void Encoder::init(MOTOR_POSITION)': lib/encoder/encoder.cpp:24:47: error: invalid use of non-static member function 'void Encoder::_itWheel()' 24 | attachInterrupt(_encoder_pins[pos], this->_itWheel, FALLING); | ~~~~^~~~~~ In file included from lib/encoder/encoder.cpp:13: lib/encoder/encoder.h:52:14: note: declared here 52 | void _itWheel();

LLM resopnse

The core problem is that `attachInterrupt` only accepts a raw function pointer with no way to pass context (no `this`). So there's no *direct* way, but here's the cleanest workaround for **4 instances**:

```
// encoder.cpp
volatile long Encoder::_wheel_counter[NUM_ENCODER_PINS] = {0};

static void (*_isrTable[NUM_ENCODER_PINS])() = {
    Encoder::_itWheelFR,
    Encoder::_itWheelFL,
    Encoder::_itWheelRR,
    Encoder::_itWheelRL
};

void Encoder::init(const MOTOR_POSITION pos) {
    _wheel_timer[pos] = 0;
    pinMode(_encoder_pins[pos], INPUT_PULLUP);
    attachInterrupt(digitalPinToInterrupt(_encoder_pins[pos]),
_isrTable[pos], FALLING);
}

void Encoder::_itWheelFR() { _wheel_counter[FRONT_RIGHT]++; }
void Encoder::_itWheelFL() { _wheel_counter[FRONT_LEFT]++; }
void Encoder::_itWheelRR() { _wheel_counter[REAR_RIGHT]++; }
void Encoder::_itWheelRL() { _wheel_counter[REAR_LEFT]++; }
```